

Performance-Focused Design Exercise Packet

Part of ASME IDETC 2025 Workshop:

Adventures in Tradespace Exploration

17 August 2025

Group ID: 5000



Annotate, mark-up, and write on this page as needed.

Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

102 in	92.3 in	76.1 in	8.6 in	715 lb-ft	4.79:1	45 in	990 kWh	0.733 in	0.28 in
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Do you think you could have found a better solution given more time?

Probably not, the main limitation was the variables themselves.

What, in this round, stopped you from finding a better solution?

- lots of info to process
- Page back not good for comparison
- vehicle changes can't be made (stuck with what we got)

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

110	80.1	65.5	20.8	421	2915:1	49	1660	0.939	1.48
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Do you think you could have found a better solution given more time?

Probably not

What, in this round, stopped you from finding a better solution?

- Very hard to find a balance
- more variation among solutions

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables

192	85.6	65.3	21.1	684	2.61:1	45	1240	0.783	1.48
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Do you think you could have found a better solution given more time?

Probably not

What, in this round, stopped you from finding a better solution?

- Scattered dots, hard to compare
- Some designs had major faults not related to graphs (energy capacity, low space, ect)

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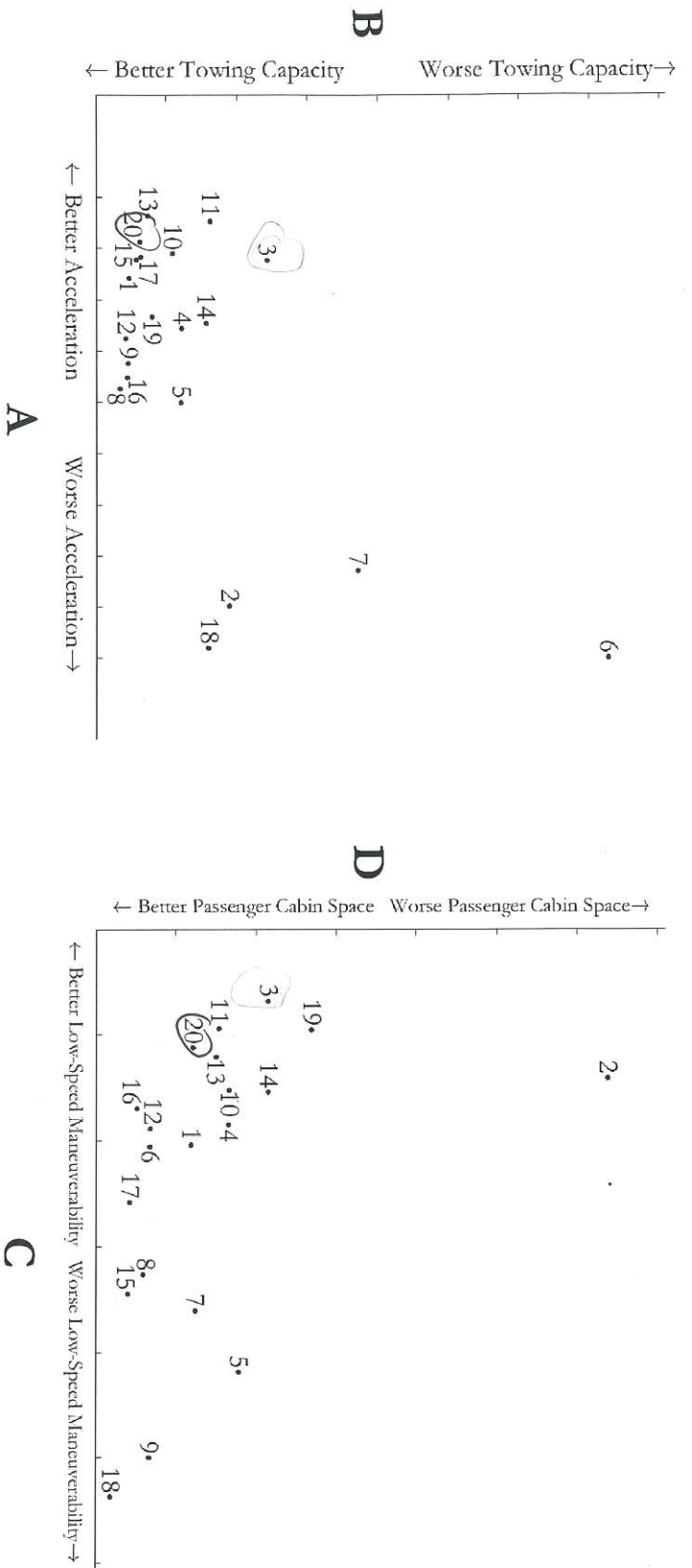
Vehicle Index

	Trackwidth	Wheelbase	Roof Height	Suspension Travel	Max Enging Torque	Driveline Gearing	Tire Diameter	Energy Capacity	Frame Rail Wall Thickness	Body Panel Thickness
1	196 in	80.2 in	79.3 in	22.9 in	728 lb-ft	5.87:1	46 in	1620 kWh	0.948 in	0.98 in
	Very Wide	Very Short	Tall Roof	Very Long Travel	Very High Output	Very Short Gearing	Very Large Tire	Very High Energy Capacity	Very Heavy-Duty Frame	Moderately Armored
2	110 in	80.1 in	65.5 in	20.8 in	421 lb-ft	2.15:1	49 in	1660 kWh	0.939 in	1.48 in
	Very Narrow	Very Short	Very Short Roof	Very Long Travel	Very Low Output	Very Long Gearing	Very Large Tire	Very High Energy Capacity	Very Heavy-Duty Frame	Heavily Armored
3	119 in	80.2 in	70.2 in	6.8 in	543 lb-ft	4.67:1	48 in	1560 kWh	0.285 in	0.18 in
	Very Narrow	Very Short	Short Roof	Very Short Travel	Moderate Output	Short Gearing	Very Large Tire	Very High Energy Capacity	Very Light-Duty Frame	Very Lightly Armored
4	181 in	81.2 in	69.2 in	9.7 in	655 lb-ft	2.37:1	32 in	1070 kWh	0.877 in	1.99 in
	Very Wide	Very Short	Short Roof	Short Travel	High Output	Very Long Gearing	Very Small Tire	Low Energy Capacity	Very Heavy-Duty Frame	Very Heavily Armored
5	192 in	85.6 in	65.3 in	21.1 in	684 lb-ft	2.61:1	45 in	1240 kWh	0.783 in	1.48 in
	Very Wide	Short	Very Short Roof	Very Long Travel	Very High Output	Very Long Gearing	Large Tire	Moderate Energy Capacity	Heavy-Duty Frame	Heavily Armored
6	178 in	81.1 in	79.1 in	6.8 in	409 lb-ft	2.53:1	47 in	1530 kWh	0.251 in	0.47 in
	Wide	Very Short	Tall Roof	Very Short Travel	Very Low Output	Very Long Gearing	Very Large Tire	Very High Energy Capacity	Very Light-Duty Frame	Very Lightly Armored
7	199 in	84.2 in	68.2 in	22.7 in	419 lb-ft	2.4:1	39 in	1520 kWh	0.331 in	1.74 in
	Very Wide	Short	Very Short Roof	Very Long Travel	Very Low Output	Very Long Gearing	Moderate Tire	High Energy Capacity	Very Light-Duty Frame	Very Heavily Armored
8	176 in	98.8 in	69.2 in	10.8 in	443 lb-ft	5.08:1	31 in	960 kWh	0.994 in	1.67 in
	Wide	Very Long	Short Roof	Short Travel	Very Low Output	Short Gearing	Very Small Tire	Very Low Energy Capacity	Very Heavy-Duty Frame	Very Heavily Armored
9	195 in	96.4 in	66.1 in	9.3 in	630 lb-ft	5.12:1	50 in	1610 kWh	0.741 in	0.17 in
	Very Wide	Very Long	Very Short Roof	Very Short Travel	High Output	Short Gearing	Very Large Tire	Very High Energy Capacity	Heavy-Duty Frame	Very Lightly Armored
10	134 in	83.1 in	80 in	18.1 in	706 lb-ft	5.95:1	49 in	1010 kWh	0.512 in	1.75 in
	Narrow	Very Short	Tall Roof	Long Travel	Very High Output	Very Short Gearing	Very Large Tire	Very Low Energy Capacity	Light-Duty Frame	Very Heavily Armored
11	107 in	87.6 in	78 in	8.1 in	750 lb-ft	2.93:1	30 in	880 kWh	0.335 in	1.97 in
	Very Narrow	Short	Tall Roof	Very Short Travel	Very High Output	Long Gearing	Very Small Tire	Very Low Energy Capacity	Very Light-Duty Frame	Very Heavily Armored
12	114 in	99.7 in	82 in	18.4 in	715 lb-ft	2.25:1	30 in	980 kWh	0.967 in	0.74 in
	Very Narrow	Very Long	Very Tall Roof	Long Travel	Very High Output	Very Long Gearing	Very Small Tire	Very Low Energy Capacity	Very Heavy-Duty Frame	Lightly Armored
13	107 in	98.2 in	66.6 in	23.1 in	512 lb-ft	5.32:1	30 in	1530 kWh	0.284 in	0.84 in
	Very Narrow	Very Long	Very Short Roof	Very Long Travel	Low Output	Very Short Gearing	Very Small Tire	High Energy Capacity	Very Light-Duty Frame	Lightly Armored
14	110 in	92 in	67.4 in	20.6 in	517 lb-ft	2.84:1	33 in	1210 kWh	0.432 in	1.82 in
	Very Narrow	Long	Very Short Roof	Very Long Travel	Low Output	Long Gearing	Very Small Tire	Moderate Energy Capacity	Light-Duty Frame	Very Heavily Armored

Vehicle Index

15	200 in	95.1 in	81.7 in	20.6 in	665 lb-ft	5.92:1	31 in	1460 kWh	0.313 in	1.67 in
	Very Wide	Long	Very Tall Roof	Very Long Travel	High Output	Very Short Gearing	Very Small Tire	High Energy Capacity	Very Light-Duty Frame	Very Heavily Armored
16	124 in	99.8 in	82.1 in	18.8 in	532 lb-ft	3.62:1	31 in	1570 kWh	0.782 in	0.33 in
	Narrow	Very Long	Very Tall Roof	Long Travel	Low Output	Moderately Gearing	Very Small Tire	Very High Energy Capacity	Heavy-Duty Frame	Very Lightly Armored
17	172 in	94 in	83.4 in	9.6 in	735 lb-ft	5.94:1	43 in	1500 kWh	0.313 in	0.78 in
	Wide	Long	Very Tall Roof	Very Short Travel	Very High Output	Very Short Gearing	Large Tire	High Energy Capacity	Very Light-Duty Frame	Lightly Armored
18	192.00527	98 in	82.4 in	6.5 in	442 lb-ft	3.23:1	35 in	1120 kWh	0.308 in	1.7 in
	Very Wide	Very Long	Very Tall Roof	Very Short Travel	Very Low Output	Long Gearing	Small Tire	Low Energy Capacity	Very Light-Duty Frame	Very Heavily Armored
19	108 in	84.8 in	66.3 in	9.6 in	456 lb-ft	5.74:1	50 in	1080 kWh	0.966 in	0.84 in
	Very Narrow	Short	Very Short Roof	Very Short Travel	Very Low Output	Very Short Gearing	Very Large Tire	Low Energy Capacity	Very Heavy-Duty Frame	Lightly Armored
20	102 in	92.3 in	76.1 in	8.6 in	715 lb-ft	4.79:1	45 in	990 kWh	0.733 in	0.28 in
	Very Narrow	Long	Moderate Height Roof	Very Short Travel	Very High Output	Short Gearing	Large Tire	Very Low Energy Capacity	Heavy-Duty Frame	Very Lightly Armored

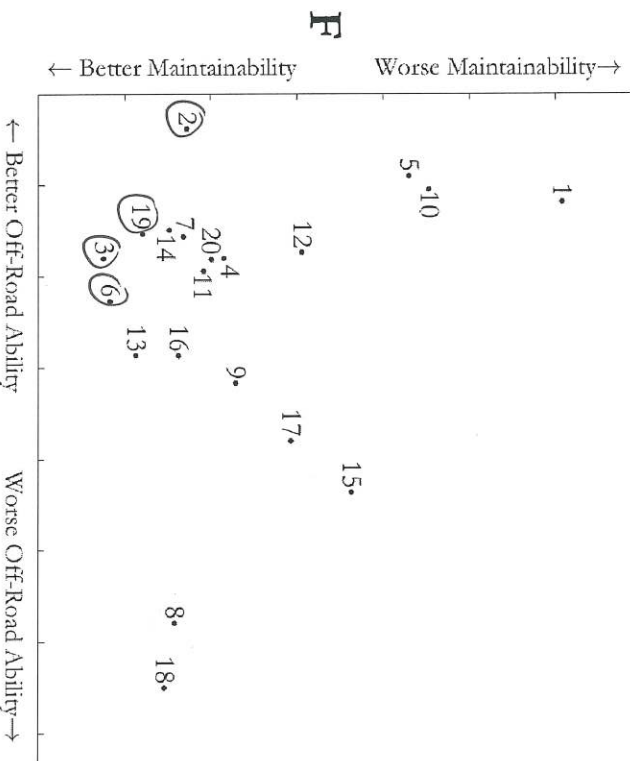
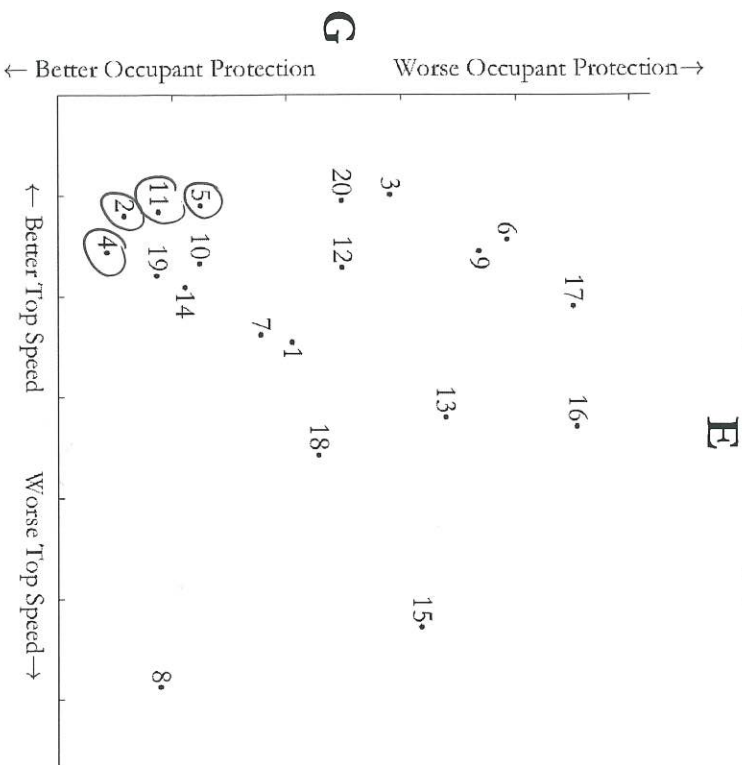
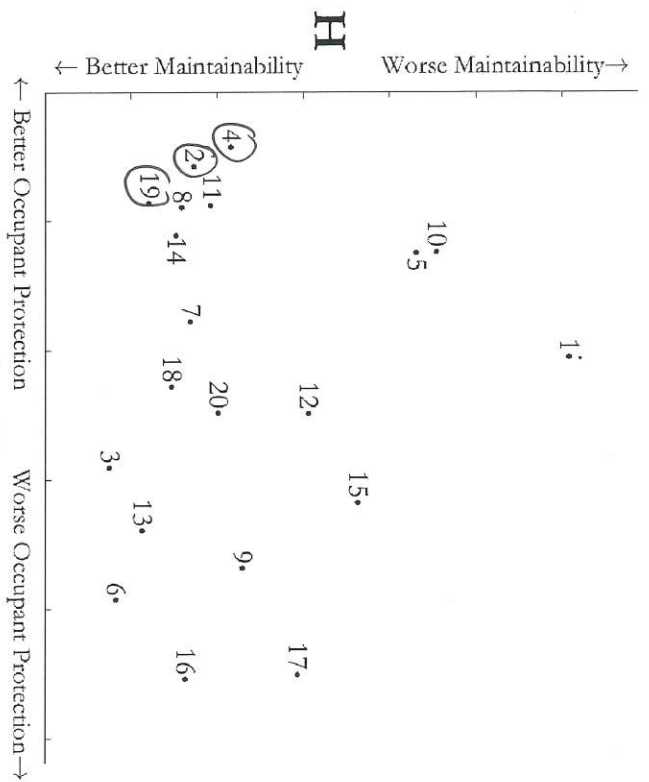
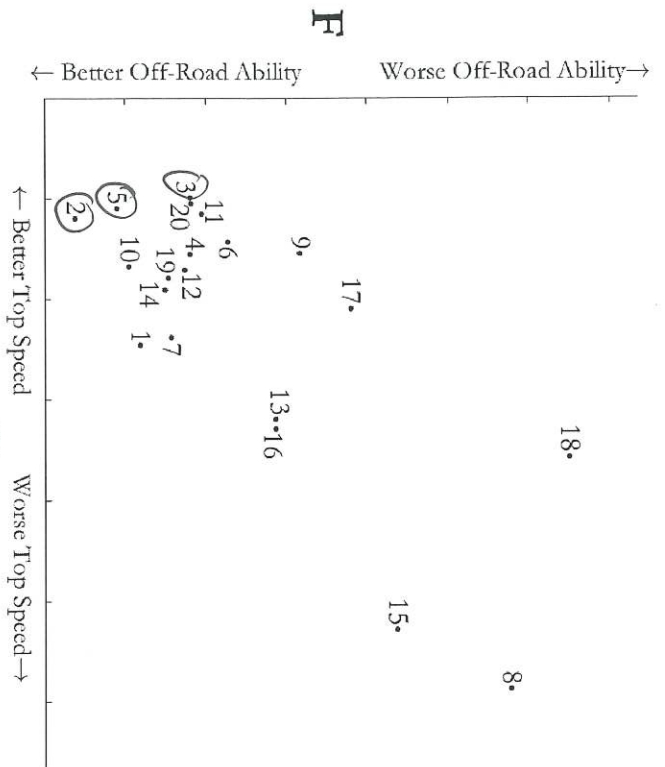
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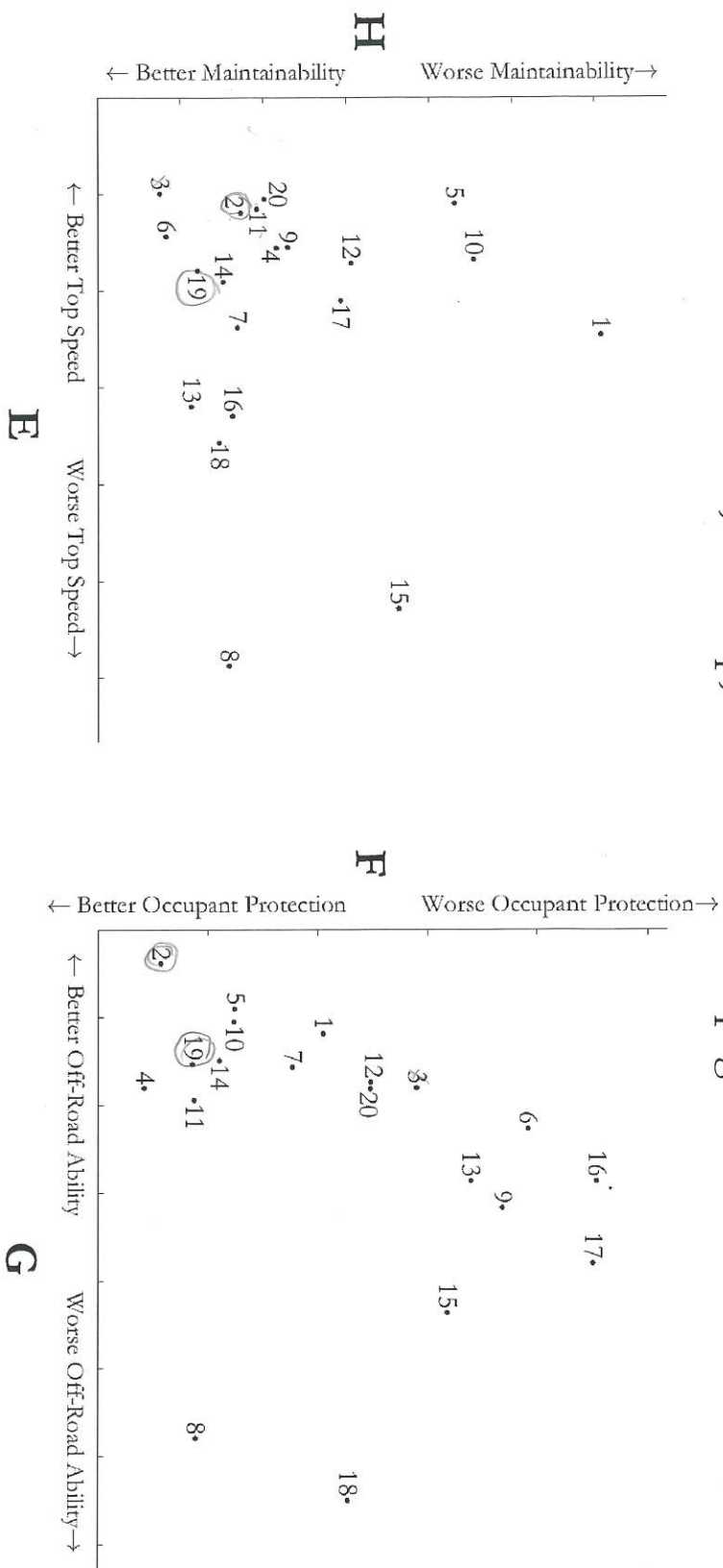
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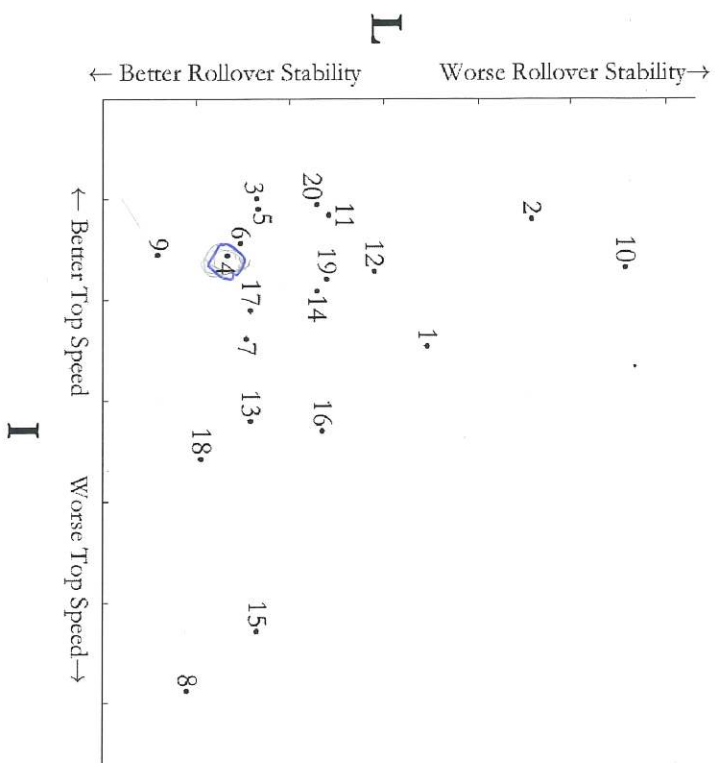
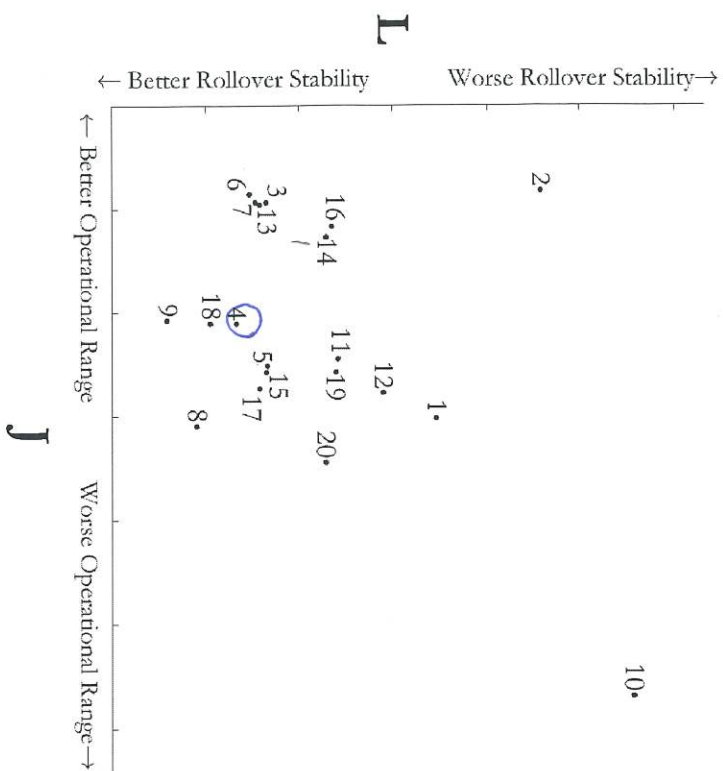
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